HS Harz FuSa, Friedrichstraße 57

# Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process an Airbag Operation

## Phase 11: Software Architecture

**Version: 1.00**

| | | |
|---|---|---|
| File | : | Airbag_ECU_Phase_11 |
| Version | : | 1.00 |
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# Contents

# 1 Meetings within the project

| Topic | Time | Room | Participants |
|---|---|---|---|
| Item Definition | 14:00 | Zoom | ALL |
| Bericht: | *none* | | |

# 2 Unresolved topics

**No unresolved topics**

# 3 Glossar

**Tabelle 1: Glossar**

| Begriff / Term | Erklaerung / Explanation |
|---|---|
| ACU | Airbag Control Unit. |
| ADC | Analog to digital converter value read from an analog sensor input. |
| ASIL | Automotive Safety Integrity Level used by ISO 26262 for risk classification. |
| CAN | Controller Area Network, a vehicle communication bus. |
| ECU | Electronic Control Unit. |
| EUC | Equipment under Control. |
| FuSa | Functional Safety. |
| HW | Hardware. |
| QM | Quality Management level below ASIL. |
| SIL | Safety Integrity Level. |
| SW | Software. |

# 4 Software Architecture

The software implementation follows a cyclic embedded control system architecture. The main loop continuously reads sensor inputs, evaluates crash logic, checks safety conditions, and sends diagnostic information to make deployment decisions. (Source: FuSa Report, Section 5)

## 4.1 System Operation Environment

(Source: ASIL Code Division V2, Section 5.1)

**Tabelle 2 System Operation Environment**

| Item | Value / Explanation |
|---|---|
| Prototype platform | Arduino Due as intended board for the report. Arduino Due has a 32-bit ARM core, 54 digital I/O pins, 12 analog inputs and CAN capability. |
| Demonstration alternative | Arduino Uno may be used in Wokwi/Tinkercad style simulation for simplified demonstration. |
| Input signals | Acceleration A0, gyroscope A1, pressure A2, passenger/baby-seat switch D2. |
| Output signals | Driver airbag indicator D8, passenger airbag indicator D9, warning lamp D13, CAN/serial diagnostic frame. |

## 4.2 Software Architecture Modules

(Source: FuSa Report, Section 3)

**Tabelle 3 Software Architecture Modules**

| Module | Function |
|---|---|
| Sensor Manager | Reads crash sensor values |
| Crash Detection Algorithm | Detects crash severity |
| Safing Logic | Independent validation logic |
| Deployment Control | Controls airbag activation |
| Diagnostics Module | Fault detection and logging |
| Communication Handler | CAN/LIN Message simulation |
| Fault Handler | Prevents unsafe operation |

## 4.3 Input and Output Signals

**Tabelle 4 Input Signals**

| Input Signal | Arduino Pin | Description |
|---|---|---|
| Acceleration sensor | A0 | Detects strong frontal acceleration |
| Gyroscope sensor | A1 | Detects angular/rollover movement |
| Pressure sensor | A2 | Detects side impact pressure |
| Passenger/baby seat switch | D2 | Disables passenger airbag |
| CAN receive | CAN RX | Receives ECU command/status |

**Tabelle 5 Output Signals**

| Output Signal | Arduino Pin | Description |
|---|---|---|
| Driver airbag indicator | D8 | Simulated driver airbag deployment |
| Passenger airbag indicator | D9 | Simulated passenger airbag deployment |
| Warning lamp | D13 | Airbag fault warning lamp |
| CAN transmit | CAN TX | Sends crash/fault status |

## 4.4 CAN Frame Design

CAN allows the airbag to send high priority crash or fault information to the main vehicle ECU.

**Tabelle 6 CAN Frame Design**

| Field | Value |
|---|---|
| CAN ID | 0x120 |
| DLC | 8 bytes |
| Byte 0 | System status |
| Byte 1 | Acceleration value |
| Byte 2 | Gyroscope value |
| Byte 3 | Pressure value |
| Byte 4 | Passenger/baby seat status |
| Byte 5 | Deployment status |
| Byte 6 | Fault code |
| Byte 7 | Simple checksum |

## 4.5 ASIL-Divided Code Structure

The code is restructured into smaller functions by safety relevance.

**Tabelle 7 ASIL Divided Function Architecture**

| ASIL | Function Name | Responsibility |
|---|---|---|
| QM | readInputsQM() | Input collection and debug support for prototype demonstration |
| ASIL D | evaluateDeploymentDecision_ASIL_D() | Highest safety relevance crash detection and safing decision |
| ASIL C | applyPassengerSuppression_ASIL_C() | Passenger airbag interlock when baby seat detected |
| ASIL D | setOutputs_ASIL_D() | Final output command for deployment indicators |
| ASIL B | checkSensorValidity_ASIL_B() | Sensor plausibility check |
| ASIL B | verifyOutputState_ASIL_B() | Output readback verification |
| ASIL B | enterSafeState_ASIL_B() | Fail-safe / fall-back reaction |
| ASIL A | sendDiagnosticFrame_ASIL_A() | Diagnostic status frame / CAN communication |
| QM | printDebugInformation_QM() | Human-readable test output only |

# 5 Figure and table listing



**No table of figures entries found.**

# 6 List of dependent documents

| No. | Dokument name | Version | Date | File name |
|---|---|---|---|---|
| 1 | Arduino Coding and Logic Implementation for Airbag ECU System | V1 | 02.06.2026 | FuSa Report_Adruino.docx |
| 2 | Airbag ECU ASIL Code Division | V02 | 02.06.2026 | Airbag_ECU_ASIL_Code_Division_V02.docx |

# 7 Literature

[1] ISO, ISO 26262-1:2011 Road vehicles - Functional safety. ISO 26262 addresses hazards caused by malfunctioning behaviour of E/E safety-related systems in road vehicles.

[2] Synopsys, What is ASIL (Automotive Safety Integrity Level)? ASIL is a risk classification system under ISO 26262; ASIL A is lowest and ASIL D is highest.

[3] Arduino Documentation, Due. Arduino Due is based on a 32-bit ARM core and provides 54 digital I/O pins, 12 analog inputs and CAN capability.

[4] Arduino Documentation, Controller Area Network (CAN) Bus. CAN is a robust vehicle bus standard for communication between microcontrollers/devices without a host computer.

[P1] Atashi Saha, Arduino Coding and Logic Implementation for Airbag ECU System, uploaded project report, 2026.

[P2] HS Harz FuSa, Airbag ECU Portfolio HS Harz FuSa - Concept template, version 1.00, 2026/04/29.

# 8 Version history

| Document name | date | Change log |
| --- | --- | --- |
| Airbag_ECU_Phase_11 | 14.06.2026 | v1.00 Initial version. Software Architecture compiled from FuSa Report Sections 3, 4, 5, 8 and ASIL Code Division V2 Sections 5.1, 5.4. |